

Saumya Kumar Saksena

PERCEPTION & EDGE AI | AUTONOMOUS SYSTEMS

☎ (+33) 758048234 | ✉ kumaar324@gmail.com | 📁 GitHub | 🗣️ Hugging Face | 🔗 LinkedIn | 🎓 Google Scholar

"You are only entitled to the action, never to its fruits." - Bhagavad Gita Chapter 2, Verse 47



Summary

Autonomy and perception engineer focused on safety-critical systems, with 5+ years of experience in real-time computer vision, and edge AI deployment. Proven track record in developing and validating perception-driven control systems under strict latency and reliability constraints.

Technical Experience

Navya Mobility

Paris, France

ROBOTICS & PERCEPTION ENGINEER

Jun. 2025 - Present

- Implement safety-layer fusion of LiDAR and Radar signals via CIP Safety protocol, combining certified detection flags to ensure robust and redundant obstacle validation independent of perception stack failures.
- Integrate and validate kinematics-aware safety zone algorithms on a new vehicle platform, performing simulation and on-vehicle testing to ensure reliable obstacle filtering during dynamic maneuvers.
- Develop safety-critical Automatic Emergency Braking (AEB) logic using multi sensor based detection, enabling fail-safe vehicle stopping within a 50 ms control loop under real-time constraints.
- Develop ASIL-B / PL-D compliant safety systems for autonomous shuttles, aligning perception-triggered control logic with functional safety requirements for deployment in regulated environments.

ams-OSRAM AG (previously 7 Sensing Software)

Paris, France

ARTIFICIAL INTELLIGENCE & COMPUTER VISION ENGINEER

Nov. 2020 - Mar. 2025

- **3D Depth Estimation:** Developed low-power DNNs for real-time 3D depth estimation via RGB + sparse dToF fusion, achieving <2% MAPE at 52 FPS (16 ms) on Samsung S20 NPU through model architecture optimization and edge deployment techniques.
- **Light Estimation:** Built TensorFlow pipeline for RGB-based local light estimation with quantization-aware training, achieving lossless 8-bit deployment and 25x latency reduction (50 ms → 2 ms) on Pixel 6 NPU without accuracy degradation.
- **Vital-signs, Respiration Rate Estimation:** Optimized and deployed low-power ML models for respiration rate estimation using PPG and accelerometer data on Cortex Mx microprocessors, with DNN-based uncertainty estimation. Also contributed to protocol design and led large-scale data acquisition involving over 40 subjects. Pending publication of 3 patents.

Aziobot B.V.

Den Bosch, The Netherlands

ROBOTICS & PERCEPTION INTERN

Jul. 2019 - Dec. 2019

- Engineered and implemented a safety layer leveraging deep learning for real-time semantic understanding, identifying 5+ flooring types with 95%+ accuracy, improving cleaning route efficiency by 20% through accurate floor-type recognition and adaptive speed control.

Indian Institute of Science

Bangalore, India

RESEARCH ASSISTANT

Jun. 2017 - May. 2018

- Designed real-time obstacle avoidance for autonomous drones in unstructured outdoor environments using dense optical flow (Farneback) and texture-based motion cues under compute constraints, achieving >90% navigation success and <50 ms reaction latency in field tests.

Education

University of Twente

Enschede, The Netherlands

MASTER OF SCIENCE IN SYSTEMS AND CONTROL

Sep. 2018 - Oct. 2020

- Relevant Courses: Image Processing and Computer Vision, 2D and 3D Scene Analysis, Advanced Computer Vision and Pattern Recognition, Image Analysis, 3D Computer Vision for Medical Analysis

BITS (Birla Institute of Technology & Science) Pilani

Hyderabad, India

BACHELOR OF ENGINEERING (HONS.) IN ELECTRONICS AND COMMUNICATION

Aug. 2013 - Jul. 2017

- Relevant Courses: Signal Processing, Image Processing, Advanced Signal Processing, Systems and Communications

Skills

Core Competencies

Perception, Autonomy, Sensor Fusion, Edge AI, Safety-Critical Systems

ML Frameworks

PyTorch, TensorFlow, TensorRT, ONNX, TensorFlow-Lite, Keras, MLflow, HuggingFace

Robotics & Simulation

ROS, MATLAB, Simulink

Programming

Python, NumPy, Pandas, OpenCV, Matplotlib, Pytest, Hydra

DevOps

Git, Jenkins (CI), Docker, Bitbucket

Tools

VSCode, Jupyter, Atlassian (Jira/Confluence), Kanban

Languages

English (native), French (intermediate), Hindi (native)